THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 65°	angles sum to 90°
2	**(d)** 17°	angles sum to 90°
3	**(a)** 62°	angles sum to 180°
4	**(c)** 40°	vertically opposite angles are equal
5	**(a)** 130°	linear pair sums to 180°
6	**(c)** 48°	corresponding angles are equal
7	**(c)** 58°	alternate angles are equal
8	**(d)** parallel lines	constant distance apart
9	**(a)** equal	formed by two intersecting lines
10	**(a)** obtuse	definition

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Parallel lines never meet however far extended (1); intersecting lines meet at a point (1).	1+1
12	Vertically opposite angles formed by two intersecting lines are equal (2).	2
13	55° (1); 145° (1).	1+1
14	Two adjacent angles on a straight line (1); they add to 180° (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	9x = 180, x = 20 (1.5); 80°, 100°, 80°, 100° (1.5).	1½+1½
16	Four angles 65° and four angles 115° (3).	3
17	3x + 10 = 5x − 30 (1.5); x = 20 (1.5).	1½+1½
	OR: The lines are parallel (1.5); converse of alternate angles property (1.5).	1½+1½
18	Place ruler along the line (1); slide set square to touch P (1); draw line along edge (1).	1+1+1
	OR: 5x + 20 = 180 (1.5); x = 32 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A line cutting two lines at distinct points (1). (ii) Corresponding, alternate, co-interior, vertically opposite (2). (iii) four of 75°, four of 105° (2).	1+2+2
	OR: (i) 68° (1). (ii) 68° (1). (iii) 112° (1). (iv) Yes, equal corresponding angles (2).	1+1+1+2
20	(i) x = 20 (2). (ii) 70° each (1). (iii) 110° (2).	2+1+2
	OR: (i) Straight angle (2). (ii) 70° and 110° (2). (iii) acute and obtuse (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 52°	angles sum to 90°
2	**(a)** 74°	angles sum to 90°
3	**(d)** 37°	angles sum to 180°
4	**(b)** 55°	vertically opposite angles are equal
5	**(d)** 68°	linear pair sums to 180°
6	**(d)** 77°	corresponding angles are equal
7	**(c)** 120°	alternate angles are equal
8	**(d)** perpendicular lines	right angle
9	**(a)** right angle	perpendicular
10	**(d)** 120°	180 − 60

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Corresponding, alternate interior, co-interior (2).	2
12	45° and 135° (2).	2
13	70°, 110°, 110° (2).	2
14	If corresponding (or alternate) angles are equal (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	180 − x = 3(90 − x) (1.5); x = 45° (1.5).	1½+1½
16	10x = 360 (1.5); x = 36° (1.5).	1½+1½
17	a + b = 180 and b + c = 180 (1.5); so a = c (1.5).	1½+1½
	OR: 90 − 70 = 20° (3).	3
18	∠BOD = 125° (1.5); ∠AOD = 55° (1.5).	1½+1½
	OR: False (1); acute + acute < 180° (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 28° (1). (ii) 62° acute (1). (iii) 90° (1). (iv) 118° (2).	1+1+1+2
	OR: (i) Steps with set square and ruler (3). (ii) Equal alternate angles measured (2).	3+2
20	(i) 3x + 20 = 5x − 40; x = 30; the angle is 110°; y = 70° (4). (ii) Vertically opposite angles equal (1).	4+1
	OR: (i) 180 − x = 5(90 − x) (1). (ii) x = 67.5° (2). (iii) 112.5 = 5 × 22.5 (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.